Convolution integrals

If we want to take the Laplace transform of the sum of functions, we can simply find the Laplace transform of each function individually, and then add the transforms to get the transform of the sum. In other words, the transform of the sum is the sum of the transforms.

For instance, to find the transform of $f(t) + g(t)$, we take the transform of $f(t)$ and $g(t)$ individually,

$$F(s) = \int_0^\infty e^{-st} f(t)\ dt$$

$$G(s) = \int_0^\infty e^{-st} g(t)\ dt$$

and then we take the sum of the transforms to get the transform of the sum. So the **transform of the sum** $f(t) + g(t)$ is given by

$$F(s) + G(s) = \int_0^\infty e^{-st} f(t)\ dt + \int_0^\infty e^{-st} g(t)\ dt$$

The product of functions

Unfortunately, we don't have the same simple process for finding the transform $F(s)G(s)$ of the product of functions $f(t)g(t)$. We *cannot* say that the transform of the product is the product of the transforms.

$$F(s)G(s) \neq \int_0^\infty e^{-st} f(t)\ dt \cdot \int_0^\infty e^{-st} g(t)\ dt$$

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Despite this inconvenient fact, it would be really helpful if we had a way of transforming the product of functions. Luckily, we do have a way of handling this, which is, of course, with the convolution integral.

Think of **the convolution** as the function we plug into the definition of the Laplace transform in order to get the product of the transforms. In other words, if we write a new function $f(t) * g(t)$ (the asterisk symbol is what we use to express the convolution of functions), then the Laplace transform of this new convolution function will give us the product of the transforms, $F(s)G(s)$.

$$F(s)G(s) = \mathscr{L}(f(t) * g(t))$$

$$F(s)G(s) = \int_0^\infty e^{-st} f(t) * g(t)\ dt$$

Of course, we can also express the convolution equation in inverse Laplace notation as

$$f(t) * g(t) = \mathscr{L}^{-1}(F(s)G(s))$$

Keep in mind that this $*$ notation does not indicate simple multiplication. So $f(t) * g(t)$ is not the same as $f(t)g(t)$; the convolution is a different thing entirely than the product. However, the convolution *is* commutative. So $f(t) * g(t)$ will be equivalent to $g(t) * f(t)$.

The convolution

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Of course, now we want to understand how to find this new convolution function, $f(t) * g(t)$, so that we can plug it into this formula to get $F(s)G(s)$. The formula we use to find **the convolution** will be

$$f(t) * g(t) = \int_0^t f(\tau)g(t-\tau)\ d\tau$$

where τ is the Greek letter "tau." And because the convolution is commutative, the formula

$$f(t) * g(t) = g(t) * f(t)$$

$$\int_0^t f(\tau)g(t-\tau)\ d\tau = \int_0^t f(t-\tau)g(\tau)\ d\tau$$

will give us the same result as the previous formula.

Let's do an example where we find the convolution of functions and show that its Laplace transform is equivalent to the product of the transforms.

Example

Find the convolution of $f(t) = t$ and $g(t) = e^t$, then show that the Laplace transform of the convolution is equivalent to the product of the individual transforms $F(s)$ and $G(s)$.

From the table of Laplace transforms, we know that the transforms of $f(t) = t$ and $g(t) = e^t$ are

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$$F(s) = \frac{1}{s^2}$$

$$G(s) = \frac{1}{s-1}$$

The product of these transforms is

$$F(s)G(s) = \frac{1}{s^2}\left(\frac{1}{s-1}\right)$$

$$F(s)G(s) = \frac{1}{s^2(s-1)}$$

So our expectation is that the Laplace transform of the convolution $f(t) * g(t)$ will give us this same result. To find the convolution, we'll first find $f(\tau)$ and $g(t-\tau)$, the two functions we need for our convolution integral.

$$f(\tau) = \tau$$

$$g(t-\tau) = e^{t-\tau}$$

So the convolution integral gives

$$f(t) * g(t) = \int_0^t f(\tau)g(t-\tau)\ d\tau$$

$$f(t) * g(t) = \int_0^t \tau e^{t-\tau}\ d\tau$$

$$f(t) * g(t) = \int_0^t \tau e^t e^{-\tau}\ d\tau$$

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$$f(t) * g(t) = e^t \int_0^t \tau e^{-\tau}\, d\tau$$

We'll use integration by parts to evaluate the integral, letting

$$u = \tau \qquad dv = e^{-\tau}\, d\tau$$

$$du = 1\, d\tau = d\tau \qquad v = -e^{-\tau}$$

Then the convolution is given by

$$f(t) * g(t) = e^t \left[uv\Big|_0^t - \int_0^t v\, du \right]$$

$$f(t) * g(t) = e^t \left[(\tau)(-e^{-\tau})\Big|_0^t - \int_0^t (-e^{-\tau})(d\tau) \right]$$

$$f(t) * g(t) = e^t \left[-\tau e^{-\tau}\Big|_0^t + \int_0^t e^{-\tau}\, d\tau \right]$$

Integrate the remaining function, and evaluate over the interval.

$$f(t) * g(t) = e^t \left[-te^{-t} - (-0e^{-0}) + \left(-e^{-\tau}\Big|_0^t \right) \right]$$

$$f(t) * g(t) = e^t[-te^{-t} - (-0e^{-0}) + (-e^{-t} - (-e^{-0}))]$$

$$f(t) * g(t) = e^t[-te^{-t} + (-e^{-t} + e^0)]$$

$$f(t) * g(t) = e^t(-te^{-t} - e^{-t} + 1)$$

$$f(t) * g(t) = -te^t e^{-t} - e^t e^{-t} + e^t$$

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$$f(t) * g(t) = -te^{t-t} - e^{t-t} + e^t$$

$$f(t) * g(t) = -t - 1 + e^t$$

$$f(t) * g(t) = e^t - t - 1$$

In theory, if we take the Laplace transform of this new function, we should get back the product of the transforms that we found earlier.

$$F(s)G(s) = \int_0^\infty e^{-st} f(t) * g(t)\ dt$$

$$F(s)G(s) = \int_0^\infty e^{-st}(e^t - t - 1)\ dt$$

$$F(s)G(s) = \int_0^\infty e^t e^{-st} - te^{-st} - e^{-st}\ dt$$

$$F(s)G(s) = \int_0^\infty e^{t-st} - te^{-st} - e^{-st}\ dt$$

$$F(s)G(s) = \int_0^\infty e^{(1-s)t} - te^{-st} - e^{-st}\ dt$$

$$F(s)G(s) = \int_0^\infty e^{(1-s)t}\ dt - \int_0^\infty te^{-st}\ dt - \int_0^\infty e^{-st}\ dt$$

Evaluate the first integral.

$$F(s)G(s) = \frac{1}{1-s}e^{(1-s)t}\Big|_0^\infty - \int_0^\infty te^{-st}\ dt - \int_0^\infty e^{-st}\ dt$$

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Evaluate the second integral using integration by parts, letting $u = t$, $du = dt$, $dv = e^{-st}\,dt$, and $v = (-1/s)e^{-st}$.

$$F(s)G(s) = \frac{1}{1-s}e^{(1-s)t}\Big|_0^\infty - \left[t\left(-\frac{1}{s}e^{-st}\right) - \int\left(-\frac{1}{s}e^{-st}\right)\,dt\right]\Big|_0^\infty - \int_0^\infty e^{-st}\,dt$$

$$F(s)G(s) = \frac{1}{1-s}e^{(1-s)t}\Big|_0^\infty - \left[-\frac{t}{s}e^{-st} + \frac{1}{s}\int e^{-st}\,dt\right]\Big|_0^\infty - \int_0^\infty e^{-st}\,dt$$

$$F(s)G(s) = \left(\frac{1}{1-s}e^{(1-s)t} + \frac{t}{s}e^{-st}\right)\Big|_0^\infty - \frac{1}{s}\int_0^\infty e^{-st}\,dt - \int_0^\infty e^{-st}\,dt$$

$$F(s)G(s) = \left(\frac{1}{1-s}e^{(1-s)t} + \frac{t}{s}e^{-st} - \frac{1}{s(-s)}e^{-st}\right)\Big|_0^\infty - \int_0^\infty e^{-st}\,dt$$

$$F(s)G(s) = \left(\frac{1}{1-s}e^{(1-s)t} + \frac{t}{s}e^{-st} + \frac{1}{s^2}e^{-st}\right)\Big|_0^\infty - \int_0^\infty e^{-st}\,dt$$

Evaluate the third integral.

$$F(s)G(s) = \left(\frac{1}{1-s}e^{(1-s)t} + \frac{t}{s}e^{-st} + \frac{1}{s^2}e^{-st} - \left(\frac{1}{-s}e^{-st}\right)\right)\Big|_0^\infty$$

$$F(s)G(s) = \left(\frac{1}{1-s}e^{(1-s)t} + \frac{t}{s}e^{-st} + \frac{1}{s^2}e^{-st} + \frac{1}{s}e^{-st}\right)\Big|_0^\infty$$

Now evaluate over the interval.

$$F(s)G(s) = \lim_{t\to\infty}\left(\frac{1}{1-s}e^{(1-s)t} + \frac{t}{s}e^{-st} + \frac{1}{s^2}e^{-st} + \frac{1}{s}e^{-st}\right)$$

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$$-\left(\frac{1}{1-s}e^{(1-s)(0)}+\frac{0}{s}e^{-s(0)}+\frac{1}{s^2}e^{-s(0)}+\frac{1}{s}e^{-s(0)}\right)$$

$$F(s)G(s)=\lim_{t\to\infty}\left(\frac{1}{1-s}e^{t-st}+\frac{t}{se^{st}}+\frac{1}{s^2e^{st}}+\frac{1}{se^{st}}\right)$$

$$-\left(\frac{1}{1-s}+\frac{1}{s^2}+\frac{1}{s}\right)$$

$$F(s)G(s)=\lim_{t\to\infty}\left(\frac{1}{1-s}e^{t}e^{-st}+\frac{t}{se^{st}}+\frac{1}{s^2e^{st}}+\frac{1}{se^{st}}\right)$$

$$-\frac{1}{1-s}-\frac{1}{s^2}-\frac{1}{s}$$

$$F(s)G(s)=\lim_{t\to\infty}\left(\frac{1}{1-s}\left(\frac{e^t}{e^{st}}\right)+\frac{1}{s}\left(\frac{t}{e^{st}}\right)+\frac{1}{s^2}\left(\frac{1}{e^{st}}\right)+\frac{1}{s}\left(\frac{1}{e^{st}}\right)\right)$$

$$-\frac{1}{1-s}-\frac{1}{s^2}-\frac{1}{s}$$

Assuming $s>0$, the denominators of all these fractions (the four that include t) will grow faster than their numerators, which means those four fractions will all converge to 0 as $t\to\infty$.

$$F(s)G(s)=\frac{1}{1-s}(0)+\frac{1}{s}(0)+\frac{1}{s^2}(0)+\frac{1}{s}(0)$$

$$-\frac{1}{1-s}-\frac{1}{s^2}-\frac{1}{s}$$

$$F(s)G(s)=-\frac{1}{1-s}-\frac{1}{s^2}-\frac{1}{s}$$

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$$F(s)G(s) = \frac{1}{-(1-s)} - \frac{1}{s^2} - \frac{1}{s}$$

$$F(s)G(s) = \frac{1}{-1+s} - \frac{1}{s^2} - \frac{1}{s}$$

$$F(s)G(s) = \frac{1}{s-1} - \frac{1}{s^2} - \frac{1}{s}$$

Find a common denominator.

$$F(s)G(s) = \frac{s^2}{s^2(s-1)} - \frac{(s-1)}{s^2(s-1)} - \frac{(s)(s-1)}{s(s)(s-1)}$$

$$F(s)G(s) = \frac{s^2-s+1}{s^2(s-1)} - \frac{s^2-s}{s^2(s-1)}$$

$$F(s)G(s) = \frac{s^2-s+1-s^2+s}{s^2(s-1)}$$

$$F(s)G(s) = \frac{1}{s^2(s-1)}$$

After all that, we found a value that matches the one we found earlier. So, at least for $f(t) = t$ and $g(t) = e^t$, we've proven that the product of the transforms is equivalent to the Laplace transform of the convolution.

$$F(s)G(s) = \int_0^\infty e^{-st} f(t) * g(t)\ dt$$

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